

Spatio-temporal Transcriptomic Profiling of Serial Human Fetal Heart Sections Reveals Cellular Architecture and Functional Niches during development

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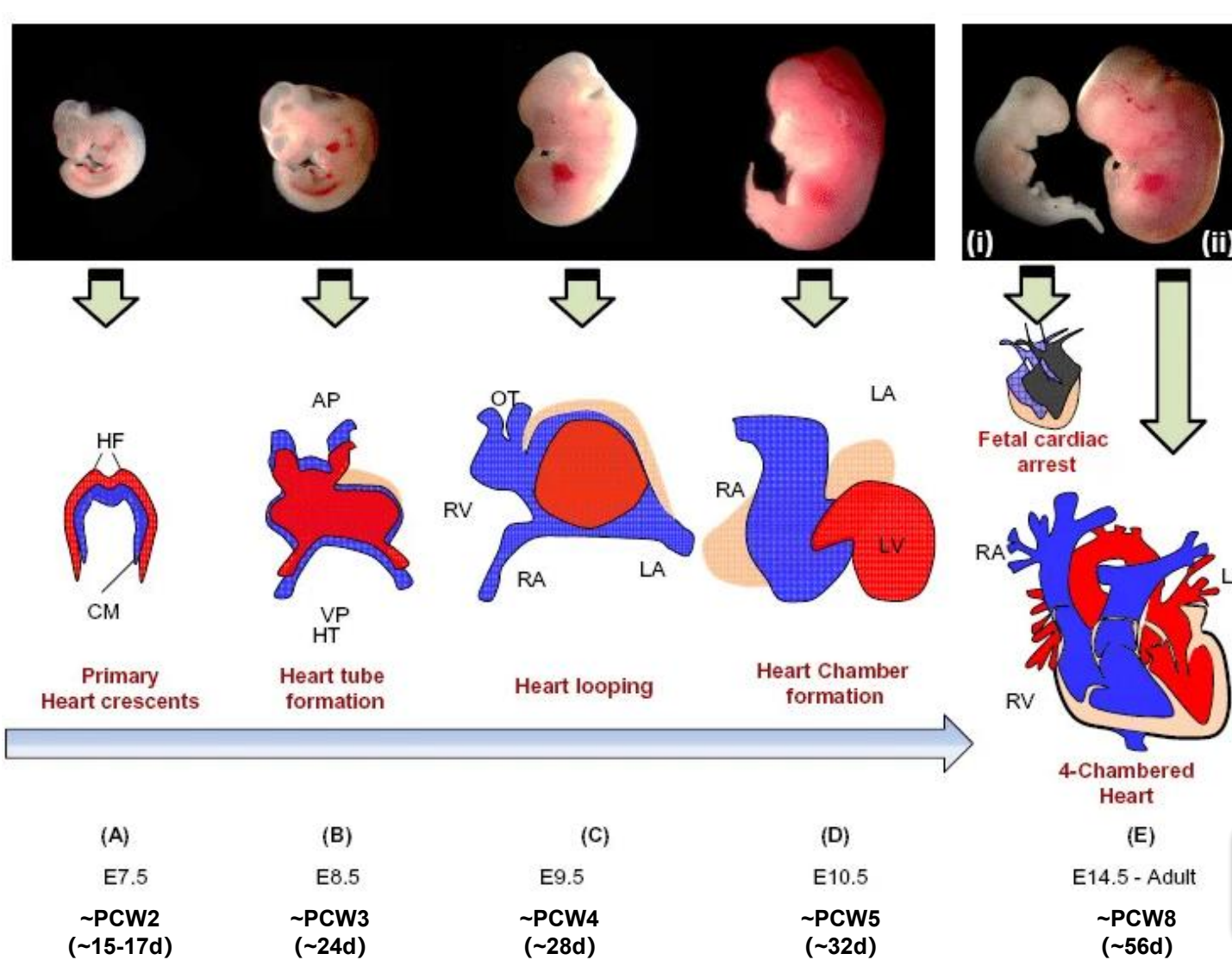
BASE INITIATIVE
Basic Science & Engineering

Abstract

Human fetal heart development requires coordinated spatial organization of cardiac, vascular, stromal, and neural populations. We utilized Illumina StrataMap Spatial Transcriptome technology to profile serial sagittal sections of the PCW7/8 human fetal heart, characterizing **cellular architecture** and **intercellular communication** while establishing a foundation for computational integration and **3D reconstruction of spatio-temporally resolved cells across various main cardiac regions**. By integrating gene-expression annotation, anatomical localization, and marker validation, we refined major cell classes into developmental subtypes, including trabecular and compact ventricular cardiomyocytes, mature ventricular cardiomyocytes, transitional conduction-like cardiomyocytes, interstitial fibroblast cells, epicardial fibroblast cells and mural cells (VSMC and pericyte). Cell-cell communication analysis revealed internal myocardial and stromal-vascular interaction hubs, strongest between interstitial fibroblast cells and mural cells, dominated by SPARC-integrin/ECM, IGFBP7-LRP1, FSTL1-DIP2A, CXCL12-CXCR4, SLIT2-ROBO, WNT-FZD/LRP and growth-factor-associated signaling axes, providing a 3D based spatio-temporal foundation for studying **early heart morphogenesis** and **predicting congenital heart disease mechanisms**.

Background

Early Human Heart Development



Critical window of early heart morphogenesis
(Weeks 2–8 post-conception)

First organ to form and function

Spatial coordination of
cardiac • vascular • stromal • neural populations

Cellular interactions & tissue morphogenesis

Normal heart development

Disruption → CHD

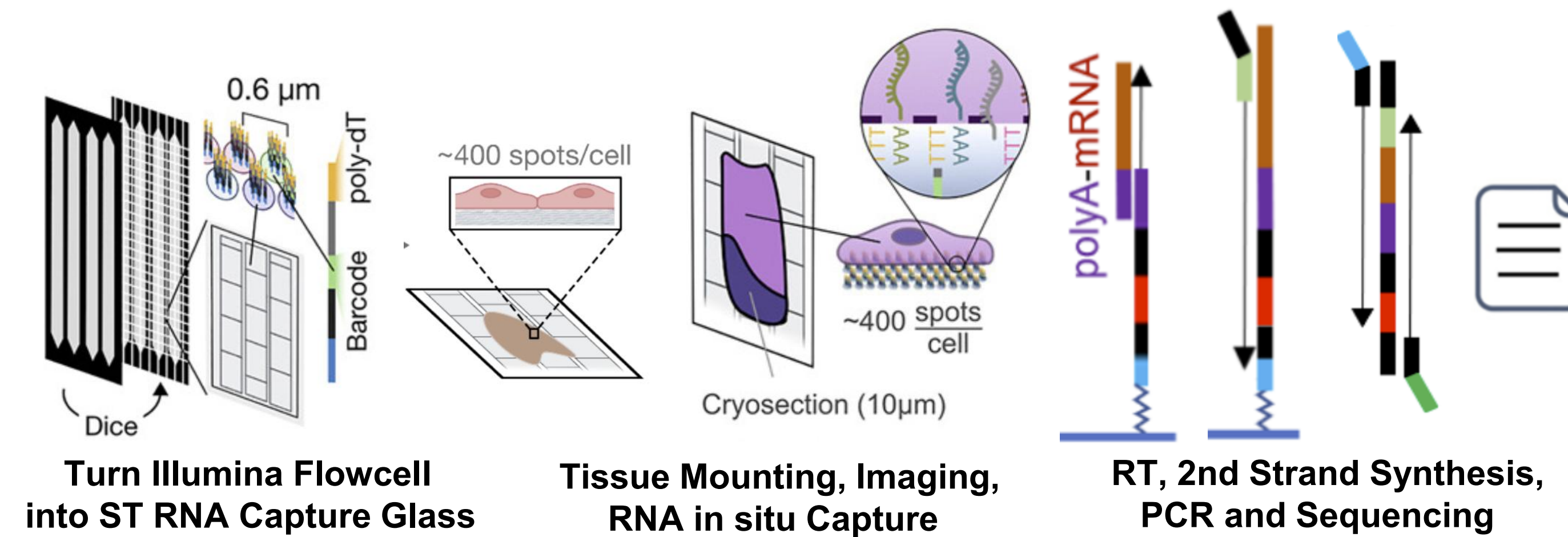
Study Objective

Construct a **spatio-temporally resolved 3D molecular atlas** of the **PCW7-8 human fetal heart** using **Illumina StrataMap Spatial Transcriptome**.

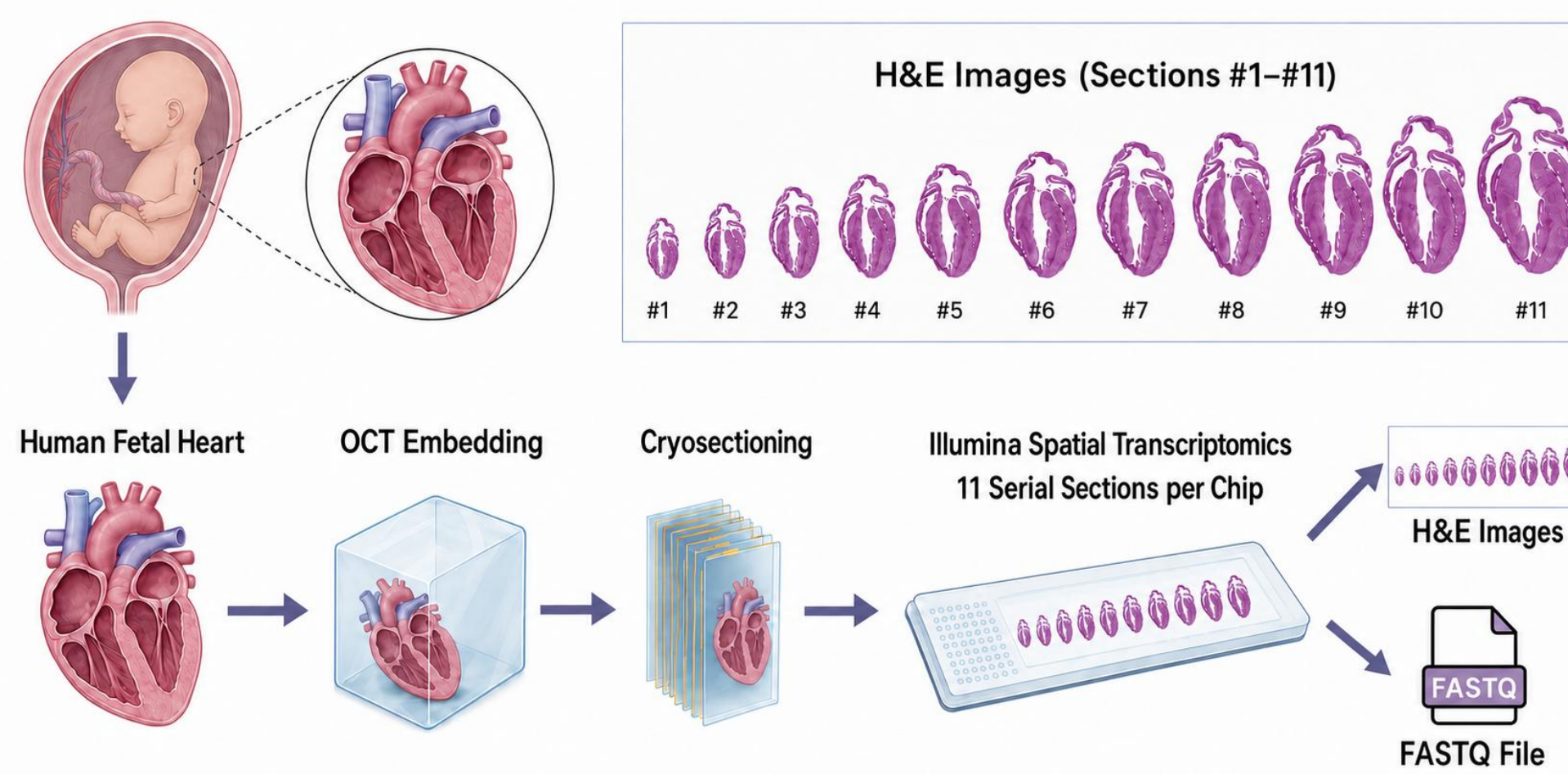
Resolve the **cellular architecture**, **intercellular communication** within niches across serial tissue sections in 3D and the **developmental trajectory** across time.

Spatial Transcriptome Profiling Workflow

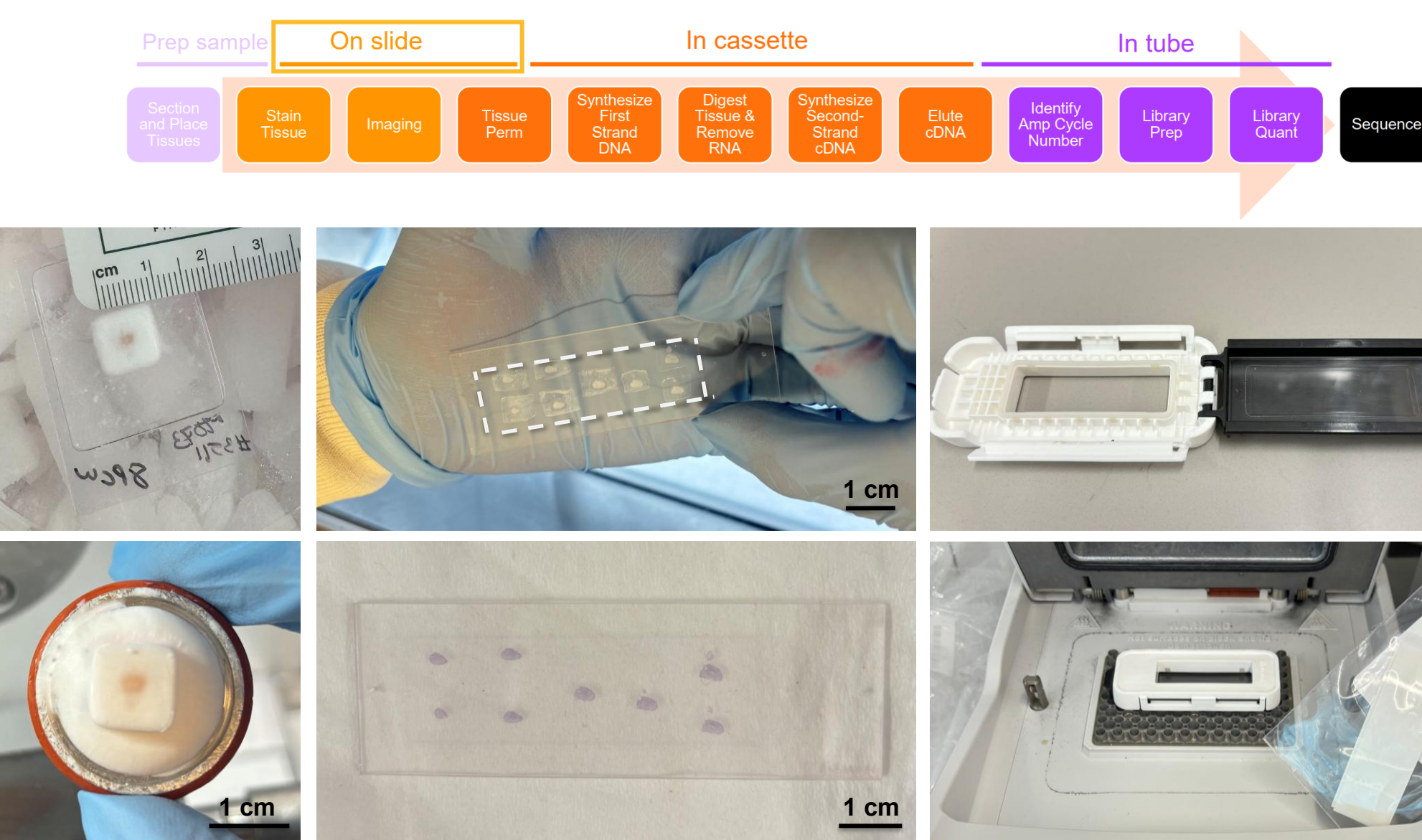
1. Illumina Flowcell Solid-Phase Chemistry: Sub-micron X-Y Location Barcoding for RNA Capture



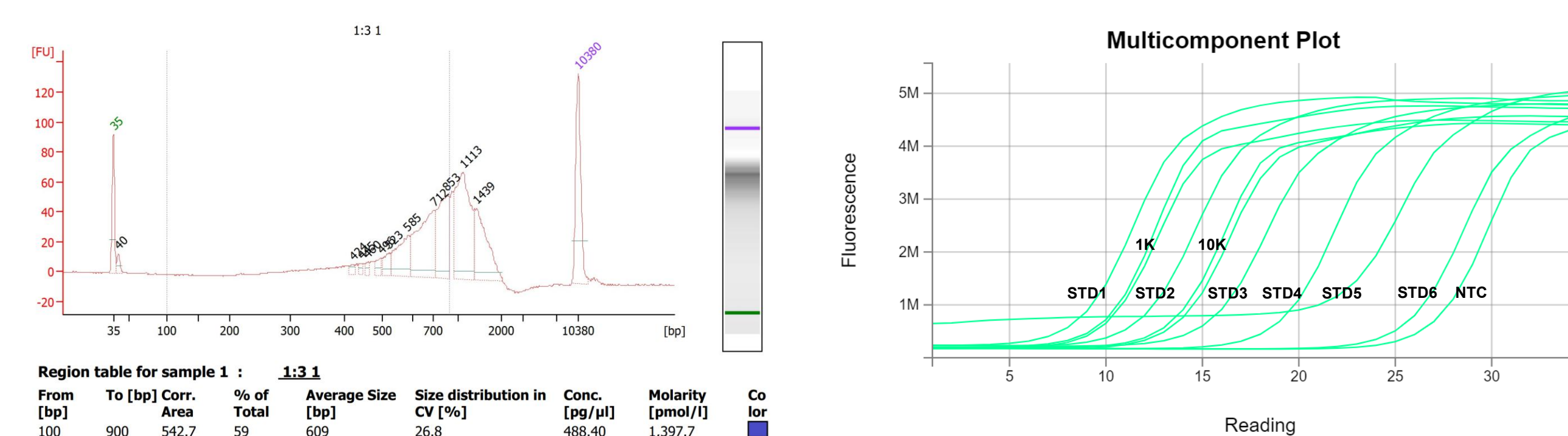
2. Overall Design of the Sample Embedding and Cryosectioning



3. Tissue Mounting, H&E Imaging, and In Situ RNA Hybridization on Barcoded Capture Glass

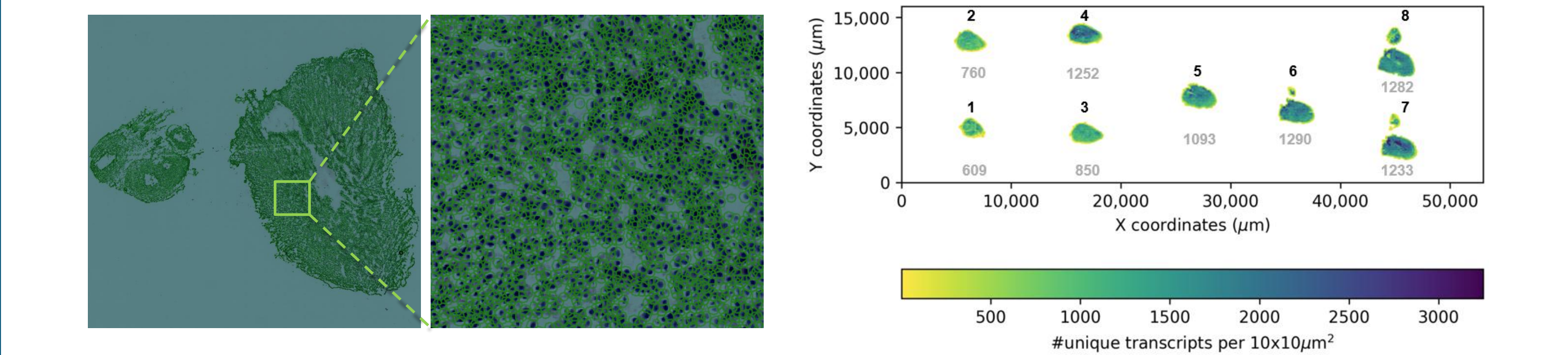


4. Barcoded cDNA Library Preparation and Sequencing

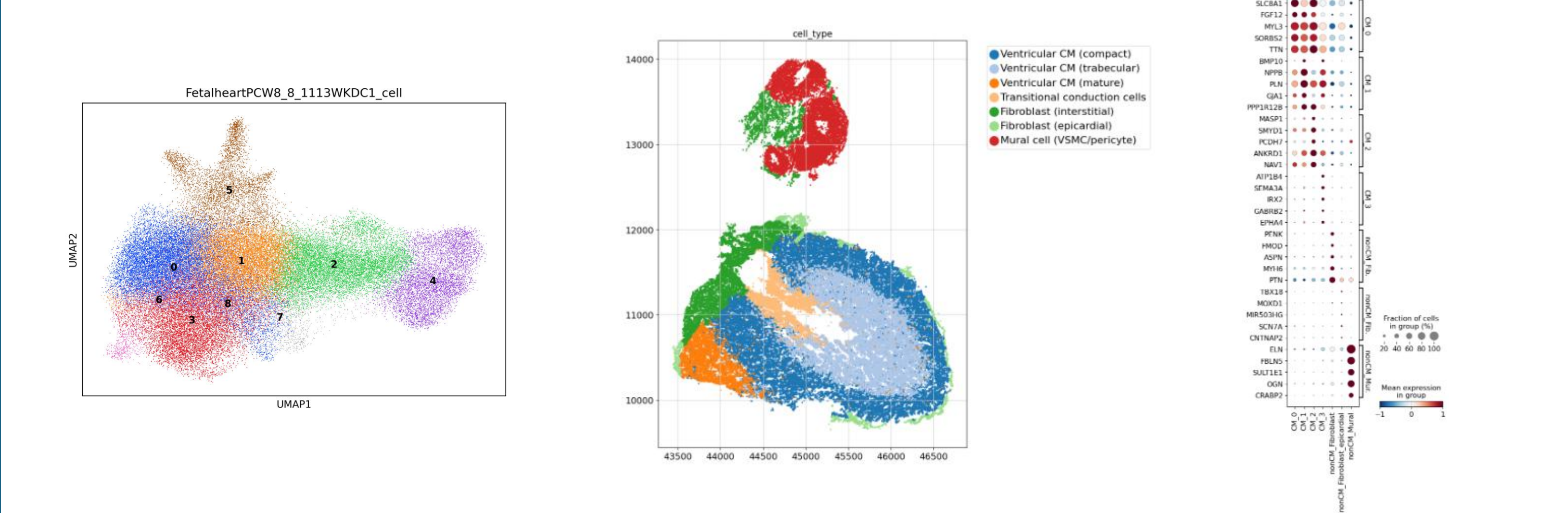


Pilot Profiling Results

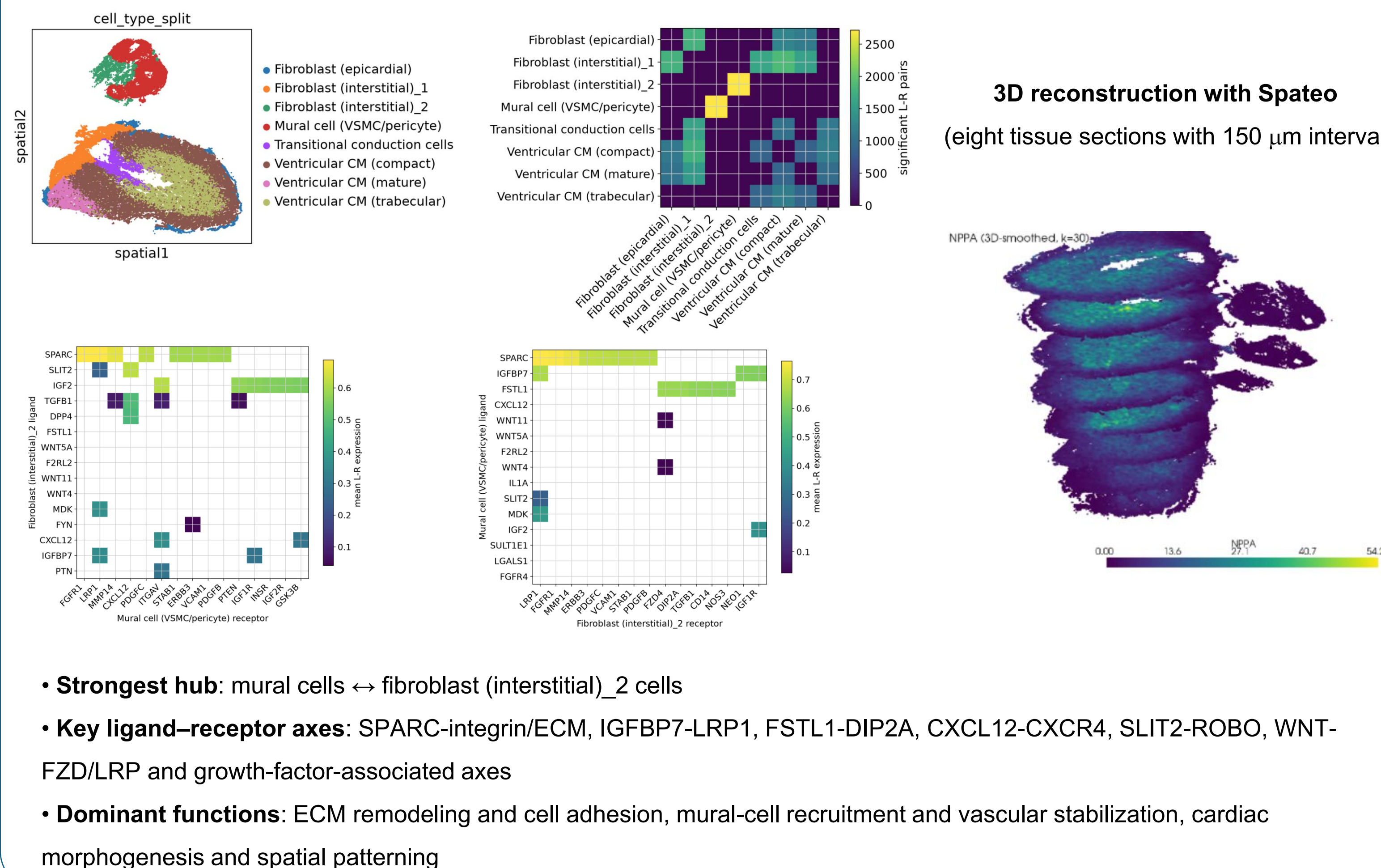
1. Cell Segmentation & Technical QC



2. Cell-Type Clustering & Annotation



3. Cell-Cell Communication Analysis and 3D reconstruction



Acknowledgement

- Illumina Company for early access of the StrataMap Spatial Transcriptome technology.
- All Qiu lab mates and collaborators for useful discussions.
- UCSF CAT core for providing the assistance in sequencing.

Reference

- Lazar et al. "Spatiotemporal gene expression and cellular dynamics of the developing human heart." Nature Genetics, 2025
- Qiu et al. "Spatiotemporal modeling of molecular holograms." Cell, 2024
- Schott et al. "Open-ST: High-resolution spatial transcriptomics in 3D." Cell, 2024
- Cho et al. "Microscopic examination of spatial transcriptome using Seq-Scope." Cell, 2021